

AMAN BHATIA

a.amanbhatia25@gmail.com | Gurugram, India | 9205287721 | [LinkedIn](#) | [Portfolio](#)

Backend-focused engineer with 4+ years of experience building scalable microservices and cloud-native infrastructure on AWS. Strong in distributed systems, event-driven architectures, and production-grade deployments using ECS, Kafka, and Terraform. Experienced in designing end-to-end systems, optimizing performance, and leading migrations from on-prem to cloud.

Education

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- Delhi Technological University (DCE), Delhi, India** | Bachelors of Technology in Software Engineering *Aug 2018 - May 2022*
- CGPA: 9.01/10.00
- International School of Beijing, Beijing, China** | International Baccalaureate (IB) (Class XII) *Aug 2017 - Apr 2018*
- IB Mathematics HL, IB Physics HL, IB Chemistry HL and IB Economics SL.

Work Experience

Senior Software Engineer | *Optum UHG (Gurugram, India)* *July 2025 - Present*

Cloud Architecture & AWS Migration

- Led end-to-end migration of **11 production services (~10TB data)** from on-prem to multi-region AWS (ECS Fargate, Aurora, S3, EFS), supporting both **real-time API ingestion and batch file-based processing workflows**.
- Designed a **multi-channel data ingestion architecture** handling: API-based ingestion via external services, file-based ingestion from shared EFS and Kafka event ingestion enabling flexible and scalable data processing across heterogeneous input formats.
- Built **routing and enrichment workflows** to process incoming data, perform identity resolution (person ID mapping), and forward enriched events to downstream systems (Kafka, Rally).
- Designed **modular processing services** (e.g., wellness/router services) to handle multiple data formats, enabling extensibility and separation of concerns across ingestion pipelines.
- Led migration, defining roadmap, mentoring 2 engineers, and owning infrastructure and deployment decisions across services.

Platform & Backend Engineering

- Owned **Kafka pipeline reliability across 20+ producers and consumers (~50K events/day)**, implementing DLQ, alerting, and replay mechanisms to ensure consistent event processing under failure conditions.
- Implemented **circuit breaking and controlled retry strategies**, improving system behavior during downstream failures and preventing overload of dependent services.
- Optimized API and database performance under peak load (~200–300 RPM), reducing **p95 latency by ~25%** and improving overall throughput.
- Migrated **14 services from Jenkins to GitHub Actions**, enabling faster and more reliable CI/CD workflows.
- Improved operational efficiency by leveraging **ECS Fargate for serverless container execution**, reducing infrastructure management overhead compared to EKS-based deployments.

Associate Software Engineer II | *Optum UHG (Gurugram, India)* *June 2022 - July 2025*

- Built backend microservices using Spring Boot and REST APIs, reducing redundant API calls and improving response efficiency in tax workflows
- Developed full-stack feature (React + Spring Boot) integrating PWC loan APIs, increasing customer adoption by **10%**.
- Designed and optimized customer portal using React, Redux, and Zustand, improving UI responsiveness (~30% improvement).

Software Intern | *Siemens (Noida, India)*

Jan 2022 – Jun 2022

- Improved log processing efficiency by **20%** using C++ Boost Serialization for structured log handling.
- Built PyQt-based data retrieval tool, eliminating manual filtering and reducing data access time.

Software Intern | *Export-Import (EXIM) Bank of India (Mumbai, India)*

May 2021 – Jun 2021

- Built a chatbot with MySQL-backed persistence, improving user adoption by 25% and automating basic query handling.

Technical Skills

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- Backend & Systems:** Java, Spring Boot, Microservices, REST APIs, Distributed Systems
 - Cloud & DevOps:** AWS (ECS Fargate, VPC, ALB/NLB, Aurora RDS, S3, EFS, IAM), Terraform, CI/CD (GitHub Actions)
 - Messaging/ Data & Frontend:** Apache Kafka, SQL, React, Redux, Zustand, JavaScript
 - Observability & Infra:** Datadog, Splunk, Kubernetes
 - Languages:** Java, Python, C++, JavaScript